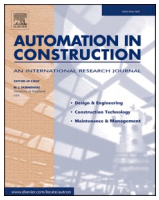
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Real-time mixed reality-based visual warning for construction workforce safety

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ABSTRACT

Spatial locations of personnel, equipment, and materials are constantly changing as construction projects progress. The dynamic nature of the construction industry affects workers' performance of identifying hazards. Even though a great deal of effort has been made to improve construction safety, the construction industry still witnesses a high accident rate. In order to complement the existing body of knowledge relating to construction safety, this paper integrates Digital Twin (DT), Deep Learning (DL), and Mixed Reality (MR) technologies into a newly developed real-time visual warning system, which enables construction workers to proactively determine their safety status and avoid accidents. Next, system tests were conducted under three quasi-on-site scenarios, and the feasibility was proven in terms of synchronising construction activities over a large area and visually representing hazard information to its users. These evidenced merits of the development testing scenarios can improve workers' risk assessment accuracy, reinforce workers' safety behaviour, and provide a new perspective for construction safety managers to analyse construction safety status.

1. Introduction

Construction safety has been considered a formidable challenge by the construction industry for a long time. According to the China state administration of work safety, the death toll in the first half of 2018 was 1752 in the China construction industry [1]. In the US, there were 1008 deaths in the construction industry in 2018, accounting for one-fifth of the recorded death toll of all industries [2]. As of now, the construction industry casualty rate still remains high.

A risk can be avoided if construction workers can promptly perceive the risk and take the correct measure. While behavioural approaches can play a part, cognitive aspects have also been an alternative approach to improve safety. There is a cognitive process from seeing a hazard to taking an appropriate measure. It will affect workers' assessment of the risk and then affect their behaviour. While the state-of-the-art extended reality research has led to pleasing outcomes in terms of technological advancement, it somewhat overlooks the profound human cognitive processes which may neutralise the efficacy of advanced visualisation applications. Studies have also shown that around 50% of the risks on construction sites are not identified promptly [3–6], and human factors, including attention, bias, risk tolerance, physical state, and mental state, can influence the visual search process used by workers to determine safety risks [7]. Dynamic and everchanging construction environment can also introduce new risks to the site. While workforce training is being used as a proactive risk mitigation measure, there are still more than 25% risks that occur due to workplace negligence, oversight or lack of awareness [8–12].

Safety management is an effective approach to enhance construction safety. The present-day management practice has been well enriched by evolving state-of-the-art information technologies [13]. For instance, an integration of Geographic Information System (GIS) and Building Information Modelling (BIM) systems can provide critical clues about hazard location, state and precaution [14–17]. Presenting these clues to onsite workers promptly can also play a part and this study aims to improve the way this information is presented. Conventionally, if someone is working and alerted of a hazard, he might have to deal with multiple tasks at the same time: continuing his task and attempting to decipher the alert. It is according to the multiple resource theory that conflicts are bound to arise when two or more tasks are performed simultaneously [18], therefore, construction workers may experience cognitive difficulties in this circumstance. Using 3-Dimensional (3D) models as information carrier is likely to be a remedy. Mixed Reality

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Table 1
Accuracy of the indoor positioning techniques.

Technique	Positioning accuracy (error)			Response time	Appearance	Reference
	Average	Median	Max			
RFID	1.25–1.70 m	1.53–1.60 m	2.95–4.00 m	Short (~30 ms)	1990s	Seco and Jiménez [40], Wu et al. [41], Seco et al. [42], Li et al. [35]
WLAN	2.06–5.30 m	2.23–4.21 m	7.5–15.0 m	Long (~1–4 s)	2000s	Zhao et al. [43], Sadhukhan et al. [44], Ali et al. [45], Li et al. [35]
Zigbee	1.15–1.62 m	/	1.64–3.00 m	Short (~60 ms)	2000s	Huang and Chan [46], Alhmiedat et al. [47], Li et al. [35]
UWB	0.13–0.24 m	/	0.23–0.51 m	Moderate (~300 ms)	2000s	Ridolfi et al. [48], Jiménez et al. [49,50], Dabove et al. [50], Cheng et al. [51], Li et al. [35]
BLE	1.59–3.80 m	/	2.55–8.00 m	Very short (~20–30 ms)	2010s	Zhao et al. [43], Zhuang et al. [52], Park et al. [38], Li et al. [35]

(MR) offers a visualisation method that can merge the virtual world with the real world. In other words, MR users can see real objects as well as 3D virtual objects. It perfectly fits the way humans observe their surrounding environment, which means that the application of MR technology could be an intuitive and practical way to present hazard information.

This paper proposes a visual warning system, which combines safety management scenarios and MR to deliver and visualise hazard information to onsite workers. The proposed system enables workers to literally see the surrounding hazardous area and the corresponding warning information in real-time using wearable MR devices. It enables workers to take proactive measures to avoid accidents. Digital Twin (DT) refers to a digital replica of a physical entity [19,20]. The ideal DT contains all the information about the physical entity and is able to synchronise any changes in real-time. Nowadays, the DT concept is gaining popularity in various application areas [21–25]. In our study, the proposed system incorporates a DT function which can simulate construction site resources and logistics. The proposed system comprises three modules: a real-time tracking module, a hazard identification module, and a visualisation module. The real-time tracking module utilises Deep Learning (DL) techniques to track the real-time location and movement trends of hazardous objects through live streaming videos taken from Closed-Circuit Television (CCTV). The hazard identification module extracts construction schedules and building geometry information from the BIM files to generate Virtual Construction Sites (VCSs). Based on the spatial context in VCSs, the module can analyse and summarise the hazard information such as hazard source, type, and location. This information is then presented on a wearable MR device via the visualisation module. Based on real-time information presented on a 3D model, workers can quickly determine the source of hazards and stay away from dangerous areas that pose a threat to them. The modules communicate with each other via sockets based on the Transmission Control Protocol (TCP).

The rest of this article is organised into five sections. Section 2 summarises techniques for hazard identification and hazard information presentation. Section 3 proposes the details of the designed visual warning system. Section 4 validates the efficacy of the presented system through preliminary experiments. Section 5 presents the results of preliminary experiments and evaluates the feasibility of the proposed system. Finally, the paper is concluded in Section 6.

2. Literature review

2.1. Hazard identification

A prerequisite of hazard information presentation is identifying hazards correctly, which requires tracking and analysing workers and objects that may cause accidents. Two categories of object tracking techniques have achieved great success in hazard identification, namely, radio-based tracking and vision-based tracking.

2.1.1. Radio-based tracking

Most commonly used tracking techniques, including Global Positioning System (GPS), Wireless Local Area Network (WLAN), Radio Frequency Identification (RFID), Bluetooth Low Energy (BLE), and Ultra-Wide Band (UWB), are radio-based [26,27]. They require the installation of signal transceiver modules on objects [14,28–32]. These techniques have already been widely used to track on-site objects, such as construction workers [33] and mobile heavy construction equipment [34]. Applying these techniques is helpful to achieve automation in the field of construction safety management [34,35]. Based on trilateration principles, triangulation principles, and scene analysis (also known as fingerprinting), the positions of objects can be determined.

For identifying hazards on construction sites, there are some issues should be considered. Li et al. [35] proposed that semi-indoor environment, complex environment, and dynamic environment are three important characteristics of construction sites that affect object tracking. They revealed that obstacles could cause the attenuation and shadowing of signals, diffraction and scattering of signals, and multipath propagation, which can affect the effectiveness of the radio-based tracking to some extent. The obstacles include construction materials, heavy equipment, vehicles, and workers. Their impact on object tracking is unavoidable since they move around constantly during construction, but the impact can be reduced through technical means. Different features of radio-based tracking techniques result in that they are applied in different situations. GPS is commonly applied for outdoor positioning since there are few overhead obstructions that can block the signal transmission between transceivers and satellites in an open area [36]. According to the research results from Pradhananga and Teizer [37], the average error of GPS devices can reach 1.1 m in an open area, and the performance of GPS degrades with increasing obstacles. As for indoor positioning, WLAN, RFID, BLE, and UWB are commonly used because their base stations (three or more fixed signal transceiver modules with known positions) can be set on construction sites. Park et al. [38] proposed a probabilistic local search algorithm for BLE. The algorithm is designed to find unreliable signals by calculating probabilities and then to estimate the positions of objects being tracked after giving the unreliable signals lower significances. Chan et al. [39] developed a transfer learning method to improve the performance of Wi-Fi fingerprinting by integrating Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM). The method can predict the positions of unlabelled fingerprints, which is helpful to reduce the accuracy drop resulted from the dynamic environment of construction sites. Table 1 summarises the accuracy and response time of the indoor positioning techniques.

Hazard identification through radio-based tracking is mainly achieved by analysing the distance between workers and hazardous areas. Jin et al. [53] proposed a monitoring system to detect, locate, and alert unauthorised intrusion for reducing safety accidents. In their research, RFID was integrated into the Internet of Things (IoT) to analyse whether a worker is entering a construction area without permission. Radio-based tracking can also track dynamic hazardous areas. Based on the analysis of the blind spot of mobile construction equipment, a proximity

warning system using RFID were developed by Teizer et al. [34] to prevent contact collisions between workers and construction equipment. Besides, Cheng et al. [54] developed a construction safety and activity monitoring application and a simulated VCS. The elements in the VCS are linked with real-time sensor data, so the application can utilise context-aware information to identify hazards. Although identifying hazards through radio-based tracking is helpful to prevent workers from injuries in a small area, equipping multiple signal transceiver modules on a real construction site is unrealistic given it is a tedious and costly practice.

2.1.2. Vision-based tracking

Vision-based tracking refers to following uniquely identified regions of interest in a sequence of images. It can track multiple on-site targets through a single camera. The development of computer vision (CV) turns vision-based tracking into a popular research topic [55]. Since the present-day DL algorithms such as CNN, Region-CNN (RCNN), Fast RCNN, and You Only Look Once (YOLO) are constantly evolving, vision-based tracking is becoming more accurate and robust [56–59]. Using models trained by DL algorithms, a computer can track on-site objects according to their features. DL has a robust capability to represent the complex relationship between input data (e.g., images) and output reasoning (e.g., object recognition). This advantage has led to reliable vision-based applications to identify and track construction workers and sources of danger [60–62]. For example, Park et al. [56], Fang et al. [57], and Fang et al. [58] used SVM, CNN and R-CNN algorithms, respectively, to estimate whether construction workers correctly use personal protective equipment. Fang et al. [63] developed a DL-based framework that can efficiently identify if workers are certified on construction sites by matching their face information and documented identities. Ding et al. [64] integrated LSTM and CNN to extract the spatial and temporal information of construction videos. The outputs can predict whether workers would climb scaffolding unsafely.

The abovementioned methods aim to track objects and identify hazards in 2D images. Such methods ignore the spatial context in the 3D environment and only cover limited hazard types. In order to address this issue, 3D image processing methods are introduced to the field of construction monitoring. Unlike 2D image processing, 3D image processing contains an additional depth map to record the distance from a camera to an object. The depth map can let a computer understand the spatial relationship between tracked objects, but getting a depth map requires extra technologies, such as structured light and time of flight [65–67], and these requirements make 3D image processing more difficult to achieve hazard identification on dynamic and complex construction sites. To this end, some researchers are trying to analyse spatial relationships between tracked objects based on 2D images. Researchers proposed two methods to obtain spatial positions. One is to form a stereoscopic vision by setting multiple cameras. It is similar to human vision in principle. If two cameras capture an object simultaneously, its spatial position can be calculated based on the camera pose (e.g., location and orientation). For example, Soltani et al. [68] proposed a monitoring approach to identify excavator poses using multiple cameras. Chi and Caldas [69] developed a safety monitoring system to identify spatial safety risks based on a stereoscopic vision. The system can detect vehicle type, pose, and velocity to evaluate whether safety rules are adhered to. Such post-processing requires considerable human input for feature extraction and computation power for mathematical calculation. Another method is to create mapping functions matching each 2D point in an image with the corresponding 3D point in the real world. Son et al. [70] proposed a function to achieve the mapping. According to the projection principle, the function matches the 2D coordinates on an image plane to the 2D coordinates on a horizontal object reference plane. The object reference plan is the plane in which the tracked objects are located, and each 2D point on the object reference plane refers to a 3D point in the real world. The mapping method was designed to provide proximity alarms to prevent collisions between workers and construction equipment in their research. It is a specific application scenario, which makes the mapping method lacks robustness to be used to improve workforce safety in more construction scenarios. On the complex construction site, a camera may capture several surfaces at different levels in an image, such as a roof surface and a ground surface. Therefore, the spatial relationship between the surfaces needs to be considered for tracking objects and identifying hazards.

2.2. Presenting hazard information

Presenting hazard information is significant for improving construction workforce safety. If workers are not given enough hazard information about their surroundings, they can hardly make appropriate decisions, which is risky. According to the human information processing model proposed by Wickens et al. [71], a worker needs to notice an accident precursor, identify it, retrieve available response options from their long-term memory, and select an appropriate response to execute. Then an accident can be avoided [72,73]. However, workers may underestimate risks or be unaware of hazards while working. Presenting information about surrounding hazards to workers is a feasible way to help them reassess the risk and keep vigilant. A traditional approach is placing warning signs in the areas where accidents may occur. The Australian Code [74] recommends setting clear warning signs, such as “Beware Explosive”, “High Voltage”, and “Danger Falling Objects” on construction sites. The approach gets workers to notice the possible hazards around them in advance so that they can take proactive actions to avoid injuries. Another traditional approach is using acoustical warnings, for example, moving forklifts can sound the horn at an intersection to alert people.

With the development of automation in construction monitoring, researchers developed several automatic approaches to present hazard information and to assist on-site workers in making optimal decisions. These automatic approaches integrate temporal and spatial information to present dynamic hazard information to on-site workers in real-time. Some researchers combined audio warning and automatic monitoring. They attach speakers to signal transceiver modules to warn workers in danger [53,75,76]. Other researchers applied tactile stimuli to warn workers [77,78]. Research revealed that the response time of human bodies to tactile stimuli is shorter than audio and visual ones [79,80]. Sakhakarmi et al. [81] developed a tactile-based wearable system to prevent collisions between workers and equipment. The system has ten vibration motors to convey different information, including the type of heavy equipment and the real-time location. Based on this, workers can make better decisions to ensure their safety. Besides, the application of advanced visualisation technology allows texts, images, and videos to be presented to on-site individuals dynamically [82,83]. Kim et al. [84] developed an innovative construction hazard avoidance system on Augmented Reality (AR) goggles. The system provides workers with hazard information, including a current safety level, the distance of the closest vehicle, and the relative location between the user and the closest vehicle. The utilise of AR allows workers to obtain hazard information intuitively [85–87]. Workers can have preliminary understandings about surrounding hazards to take proactive actions to avoid accidents. Their system makes effective use of the focal vision channel. However, considering construction workers’ daily tasks may occupy their focal vision channel, conflicts may occur [88,89]. Therefore, using ambient vision to provide warnings might be a great option while construction workers are working [71].

2.3. Research gaps and objectives

The reviewed research indicates that many studies on the topic of construction safety heavily rely on radio-based or vision-based technologies. However, radio-based tracking methods require the installation of signal transceivers on the objects to be tracked. Equipping multiple signal transceiver modules on each construction site is

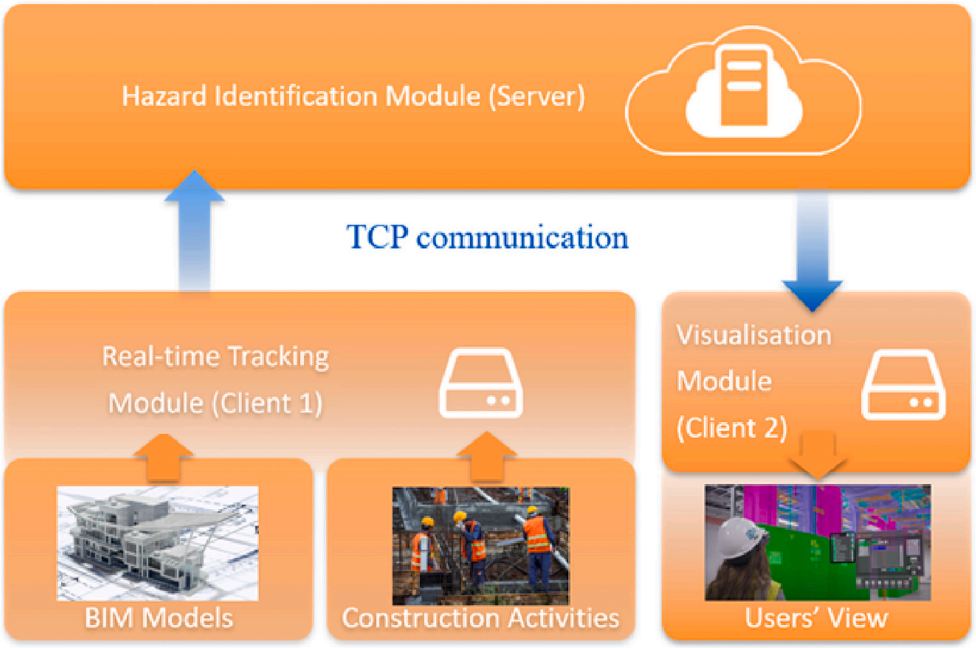


Fig. 1. The proposed visual warning system architecture.

unrealistic, given it is a tedious and costly practice. Therefore, this type of method is not the scope of our study. Regarding vision-based methods, they are hard to define safety risks in different scenarios. To be specific, some construction resources or workers' behaviours are considered dangerous in specific spatial contexts. For example, using a welding gun near acetylene or oxygen cylinders is riskier than in other places since it can cause an explosion. Using safety harnesses is mandatory when working at height, but it is unnecessary when working on the ground. Although many researchers have done excellent works to achieve the representation of complex construction safety knowledge through ontology modelling [90–93], the integration of ontology and CV for hazard identification in 3D spaces has not been fully achieved yet [94]. The real-time reconstruction of dynamic 3D construction sites through CV alone is still a difficult task at this stage, which challenges workers to identify spatial-related hazards intuitively. To this end, this study integrates DL and DT, which can determine the spatial location of tracked objects based on the geometric information of the BIM. This integration can not only provide spatial context information but also obtain the spatial relationship between objects. By inputting and updating BIM models, hazards can be found in the corresponding environment, increasing the robustness of hazard identification.

Additionally, many hazard warning methods have been developed through audio and tactile ways. But this type of warning simply aims to make workers aware of danger without providing additional safety information. It is also difficult for workers to quickly determine their safety status. To this end, some studies provide workers with additional safety information to help workers make decisions and proactively avoid hazards [84]. Research has shown that AR and MR techniques merging virtual objects with the real world can significantly improve workers' safety awareness [95,96]. However, hazards are mainly shown in workers' focal view in these studies. The displays of hazards do not conform to the human habit of identifying hazards. Because the construction tasks have already taken up workers' focal vision. Delivering safety information in their focal view can reduce their working performance. To the best of the authors' knowledge, there is a lack of a spatial context-aware method to provide real-time safety information for on-site workers in their ambient vision. The integration of DT and MR can deal with the issue and provide visual warnings to on-site workers through both focal and ambient vision channels. The purpose of this study is to bridge the research gap mentioned above by integrating multiple advanced technologies. To achieve this purpose, two objectives are pursued: a) integrating DL and DT to increase the robustness of hazard identification, and b) integrating DT and MR to generate and provide real-time safety information.

3. Methodology

The proposed visual warning system focuses on proximity alerting and it consists of three modules. These modules are responsible for real-time tracking, hazard identification, and visualisation, respectively, as shown in Fig. 1. The real-time camera tracking module can do object tracking and calibrate the coordinates. In the hazard identification module, information from 4-dimensional (4D) BIM models is used to create a VCS. By inputting the information obtained from the real-time tracking module into the hazard identification module, the objects can be reconstructed as DTs in the VCS. Then, according to safety regulations, hazard information can be generated, including hazard sources, hazard types, and hazardous areas. The visualisation module can present the hazard information to a worker through an MR goggle when he enters a hazardous area. These modules communicate with each other through TCP sockets. In order to reduce the impact of network latencies, each module has its own processing units to ensure that the transmitted data is as simple as possible.

3.1. Real-time tracking module

The real-time tracking module combines the YOLO version 4 Tiny (YOLOv4-Tiny) algorithm [97] with Deep SORT [98] to detect and track safety-related objects in live streaming acquired from a CCTV camera on a construction site.

YOLOv4-tiny is a lightweight version of the 4th generation of YOLO that requires less computational resources and memory and has high throughput. This characteristic makes it suitable for processing live streaming. YOLOv4-tiny consists of three main modules, which are CSPDarknet53-Tiny, Feature Pyramid Network (FPN) and Yolo Head. CSPDarknet53-Tiny is developed for feature extraction of input images. It comprises Convolutional layers (Conv) and CSPBlock. The Leaky ReLU (The Rectified Linear Unit) function is employed as the activation

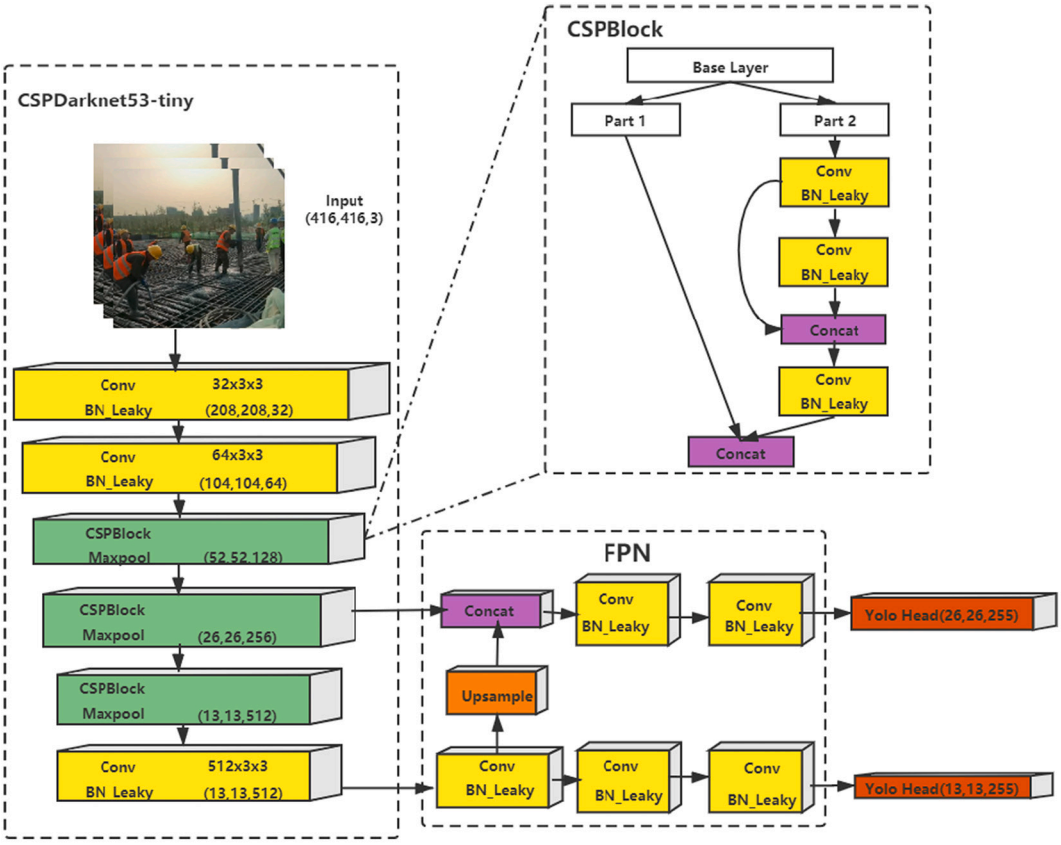


Fig. 2. The YOLOv4-tiny overall architecture.

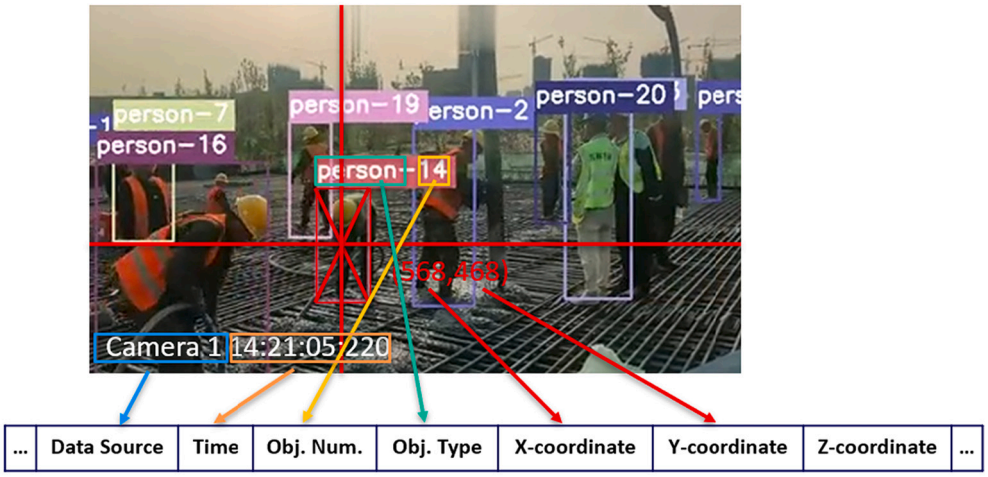


Fig. 3. Encoding information into a byte array.

function in Conv to enhance the processing speed. The structure of CSPBlock can reduce the computational complexity without compromising the detection performance. FPN is designed to produce features maps by fusing features from different network layers, which can avoid losing the semantic and geometric information of images. Yolo Head utilises the feature outputs for the final prediction of classes. The overall architecture of YOLOv4-tiny is shown in Fig. 2. In this study, the pre-trained weights of YOLOv4-tiny were implemented to perform object detection tasks. These weights were trained and evaluated on the MS-COCO dataset [99], containing 80 general classes. Since the MS-COCO dataset does not contain safety-related objects on construction sites (e.g., excavators, safety helmets, safety harness), only the "person" and "truck" classes were considered for object detection tasks in this study. In the future, custom datasets that include multiple safety-related objects on construction sites will be developed to train the object detectors.

Deep SORT was used to conduct the tracking procedure based on the outputs of YOLOv4-tiny in each frame. It generates unique IDs for each detected object. In order to improve tracking accuracy and efficiency, Deep SORT was integrated with Kalman filter [100] and Hungarian algorithm [101]. With the help of these two algorithms, frame-by-frame data is associated and correlated, and the positions of detected objects are tracked based on their positions in previous frames. After that, the

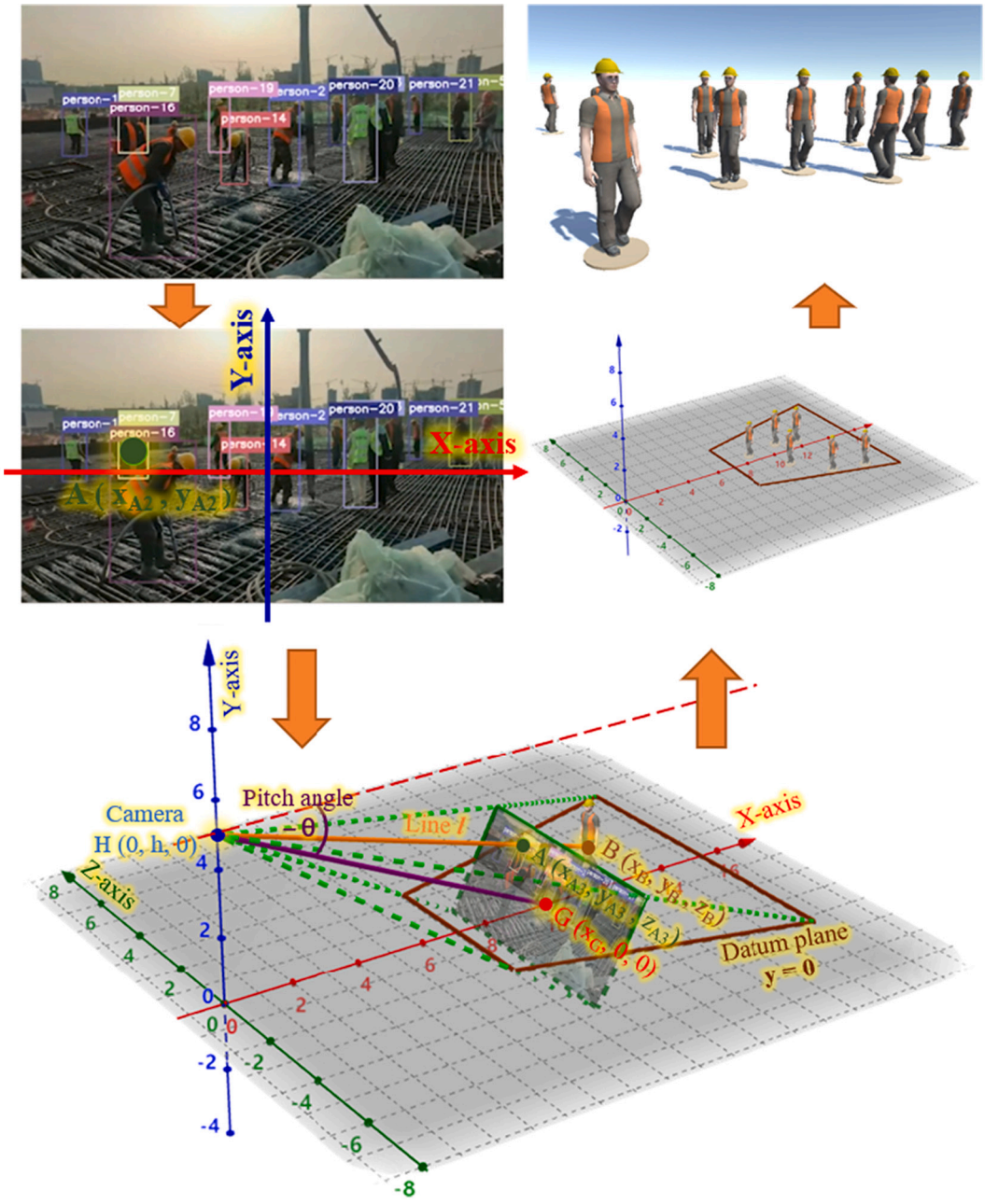


Fig. 4. Construction worker digital twins.

IDs and classes of the tracked objects and their 2D coordinates in the frame are transmitted to the hazard identification module via TCP sockets in real-time.

In addition, the technology for tracking MR devices is Simultaneous Localisation and Mapping (SLAM). SLAM is created to generate a map of surroundings and locate an agent based on image sequences taken by the agent [102]. Since a virtual world with hazard information is to be merged into the real world, the position and orientation of MR devices must be very accurate, which can be achieved by the SLAM. It is worth noting that some MR devices, including Microsoft HoloLens being used in this study, feature SLAM technology to help the devices understand their surroundings [103]. Therefore, SLAM can be applied directly to the system via Microsoft Application Programming Interface (API).

3.2. TCP communication

Since the proposed system may include multiple real-time tracking modules and visualisation modules, it was designed based on a server-client architecture. The hazard identification module is the server, and the other modules are the clients. Sockets for TCP communication link the server with the clients. The server assigns address information such as IP address and port number to its socket. Clients can connect their socket to the server socket by accessing the server address. Then, the information can be transmitted between the server and the clients through TCP communication after its carriers are encoded into byte arrays. In order to reduce the impact of network latencies, the information carriers are numerical data and texts rather than images. As shown in Fig. 3, data source, time, object number, object type, x-coordinate, and y-coordinate are encoded in sequence when information is transmitted from a real-time tracking module to a hazard identification

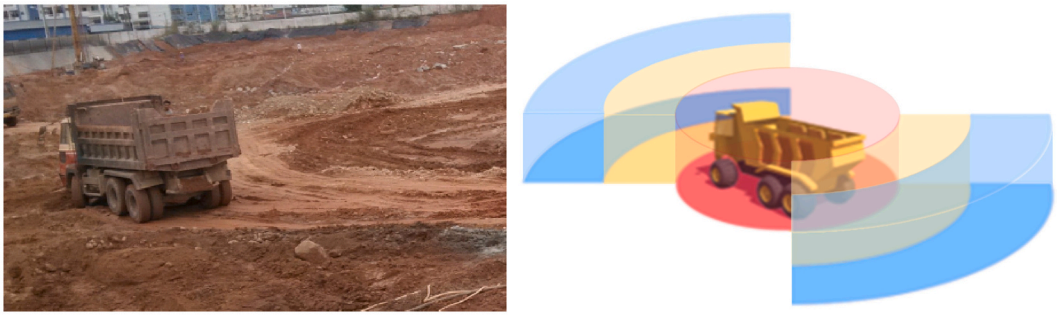


Fig. 5. The colour-coded hazardous area near a truck.

module. When information is transmitted from a hazard identification module to a visualisation module, a z-coordinate is also encoded and transferred.

3.3. Hazard identification module

The function of the hazard identification module is to integrate and analyse information obtained from different sources to identify hazards. A VCS was created in the module to simulate construction activities based on real-time information. It was developed using the Unity game engine since the engine has good extensibility and rich community resources [104,105]. A VCS can obtain 3D models of a construction project from 4D BIM files by imputing OBJ format files manually or using synchronisation plug-ins, such as Pixyz and Unity Reflect. The schedules in the 4D BIM can be used to set the appearance time of building components. As the server of the proposed visual warning system, the hazard identification module gathers information from all real-time tracking modules. Each area monitored by a real-time tracking module is linked to an area in a VCS. When information of a tracked object is transmitted to the hazard identification module, a DL of the object is generated in a VCS through a mapping approach.

The mapping approach is created based on the research results from Son et al. [70]. A 2D image taken by a camera is a projection of a 3D space onto a plane. Based on the camera pose and the object coordinates in the image, the object position in the real world can be determined. A coordinate system and several formulae were designed to achieve the mapping.

As shown in Fig. 4, the camera is located at point H (0,*h*,0), and its pitch angle is –θ. The datum plane is the plane on which the object is located. The datum plane should be set on the same level as the tracked object's centre (y = 0 in Fig. 4). The line HG is the camera's optical axis, and the point G (x_G, 0, 0) is the point of intersection of the optical axis and the datum plane, thus,

$$x_G = h/tan\theta \tag{1}$$

According to the perspective projection principle, a captured image is perpendicular to the optical axis of the camera lens, and its projection onto the datum plane is a trapezoid. To simplify the model, we set the midpoint of the image at point G. For any 2D point A (x_{A2}, y_{A2}) in the image, its 3D coordinates are (x_{A3}, y_{A3}, z_{A3}) in the 3D coordinate system. According to the geometric relationship,

$$\begin{cases} x_{A3} = x_G - y_{A2}{}^{*}sin\theta \\ y_{A3} = -y_{A2}{}^{*}cos\theta \\ z_{A3} = -x_{A2} \end{cases} \tag{2}$$

The equation of the line *l* that passes through the point A and H is

$$\frac{x}{x_{A3}} = \frac{y-h}{y_{A3}-h} = \frac{z}{z_{A3}} \tag{3}$$

The coordinates of point B where the line *l* intersects the datum plane are (x_B, y_B, z_B). Thus,

$$\begin{cases} x_B = \dfrac{h}{h-y_{A3}}{}^{*}x_{A3} \\ y_B = 0 \\ z_B = \dfrac{h}{h-y_{A3}}{}^{*}z_{A3} \end{cases} \tag{4}$$

Therefore, any 2D point in an image can be transferred into a 3D point. Then, the coordinate system can be integrated into a world coordinate system in a VCS by multiplying a transformation matrix.

By putting 3D models of the tracked objects at the corresponding 3D points, the movements of the objects can be synchronised in the VCS. Each 3D model is a DT of the tracked object. Therefore, a DT's position and movement in the VCS reflect the position and movement of the corresponding real object on a construction site. The VCS in the hazard identification module has information from 4D BIM so that it can provide context-aware information. Besides, objects at different heights can be tracked accurately by generating different datum planes based on different surfaces on which objects are attached, such as ground surfaces, floors, and ceilings. The proposed mapping approach achieves tracking multiple objects on surfaces at different heights, which increase the robustness of hazard identification using 2D image-based tracking. The distance between a tracked object and a system user is the key to estimating risks. In the system, red, yellow, and blue zones are designed to indicate different risk levels of hazardous areas. Each hazardous area is associated with an object that may cause an accident. The red zone is where the accident may occur at any time; the yellow zone is the area that may become a red zone in the next few seconds, and the blue zone is the area that may become a yellow zone in the next few seconds. For example, the area adjacent to a truck within 3 m is the red zone, as shown in Fig. 5, since a truck may suddenly overturn, move forward or move backward. Workers in this area can be injured at any time. The sectors in the front and rear of a truck are yellow and blue zones. Since the speed of a large vehicle on the construction site must not exceed 5 km/h, the yellow zone is within 10 m, and the blue zone is within 17 m when assuming workers needs 5 s to get out of the hazardous areas. If a system user enters a hazardous area, the hazard identification module determines that the user's safety is being threatened. Then the corresponding hazard information will be generated and sent to the visualisation module through TCP immediately. It is worth noting that the latency and the speed of tracked objects can affect the identification of dangerous areas. Thus, a script was created to automatically move the generated hazardous areas forward in the direction of the object's movement according to the speed and latency.

3.4. Visualisation module and warnings

The visualisation module is to present information about hazards around system users. MR goggles are selected to achieve visualisation since they can merge a virtual world to the real world, allowing workers to obtain additional information while viewing their surroundings. The



Fig. 6. The explosion warning representations: (a) with a virtual avatar overlay (b) without an overlay.

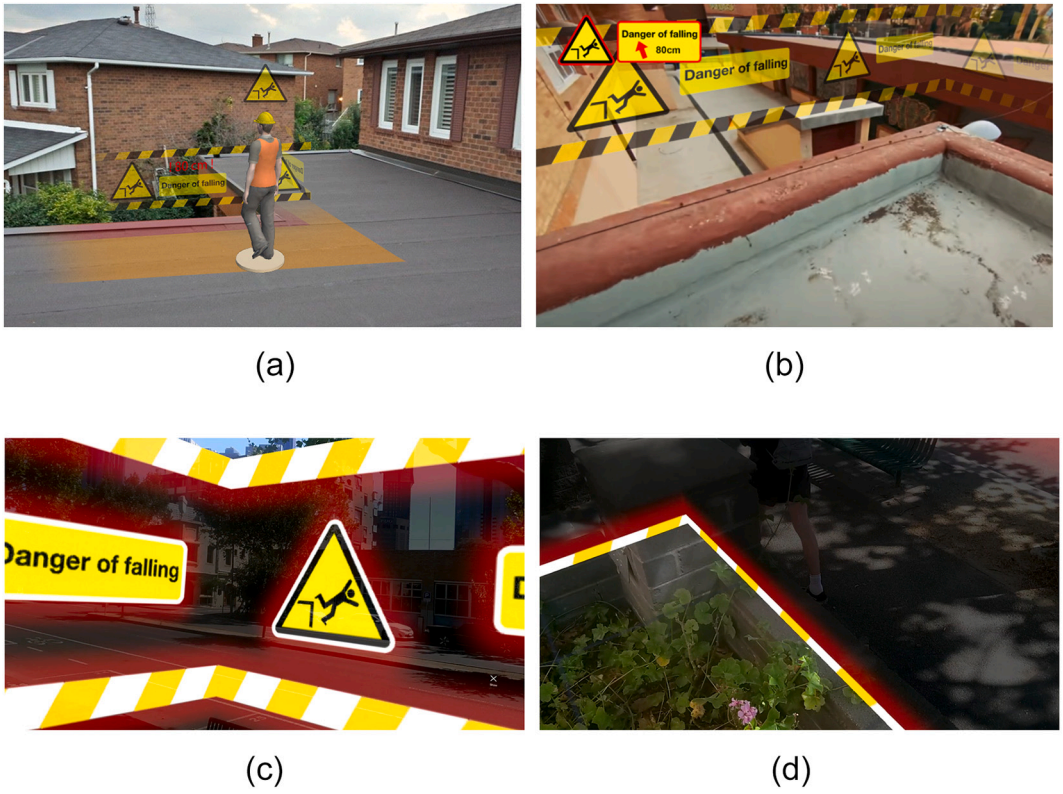


Fig. 7. The falling from height warning representations: (a) concept third-person view (b) concept first-person view (c) system user's view (d) alignment between virtual and real worlds.

Unity engine is also suitable for the development of MR applications. Therefore, a simplified VCS was developed for the visualisation module. It contains several positioning anchors, information display windows, and a series of pre-defined 3D models of dangerous objects, such as cars, trucks, excavators, and gas cylinders.

The hazard information transmitted from the hazard identification module can be displayed on the windows, which utilises workers' focal vision to provide warnings. Besides, the 3D models are used to warn workers through ambient vision. The models visualise the hazard sources and hazardous areas intuitively. Since the proposed system aims to enhance construction safety from the perspective of proximity warning, several warnings were designed to help construction workers



Fig. 8. The struck-by accidents warning representations.

Table 2
Development environment of the proposed system.

Component	Server computer	Client computer
CPU	Intel Core i5–7500	AMD Ryzen 7 3700X
RAM	16 GB	16 GB
Graphics card	NVIDIA GeForce GTX 1060	NVIDIA GeForce GTX 1660
System	64-bit Microsoft Windows 10	64-bit Microsoft Windows 10
Python version	/	Python 3.6.5

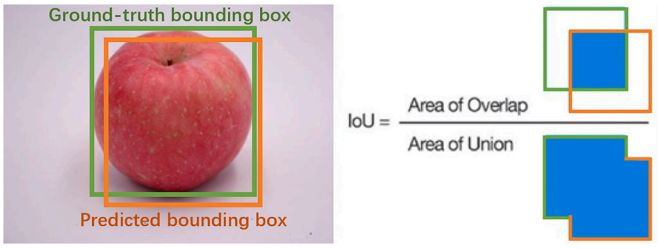


Fig. 9. An example of IoU.

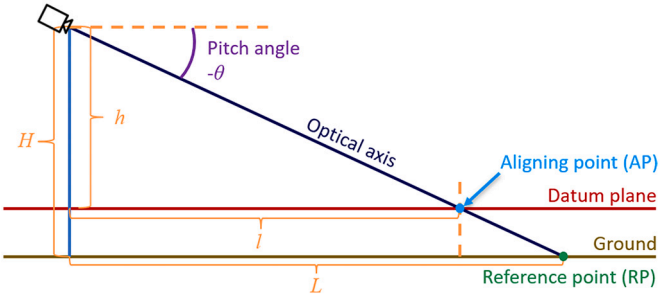


Fig. 10. The camera position and orientation.

keep alert and avoid injuries. Three kinds of hazards, including explosion (Fig. 6), falling from height (Fig. 7), and struck-by accidents (Fig. 8), were selected as sample scenarios. If a worker is in a blue zone, a warning sign is shown as a 'hologram' to notice the worker about the hazard. If a worker is in a yellow zone, the red zones will be shown to help him keep away from the hazard. If the hazard source is not in the user's view, a 3D arrow will be displayed to direct the user's eyes to the hazard source. If a worker is in a red zone, a voice assistant and a flashing exclamation mark will warn him to leave the red zone.

Registration is a vital issue in MR, which refers to the alignment of a virtual world to the real world. In the proposed system, the registration

Table 3
Summary of experiment settings.

Equipment	
Camera	iPhone 12 Mini camera
MR device	Microsoft HoloLens
Network	Hotspot (iPhone 12 Mini)
Scenarios 1 & 2	
Height of datum plane (h)	0.88 m
Height of camera (H)	5.68 m
Pitch angle ($-\theta$)	-25 °
The horizontal distance between AP and camera (l)	10.30 m
The horizontal distance between RP and camera (L)	12.19 m
Scenario 3	
Height of datum plane for the system user (h_1)	0.98 m
Height of datum plane for cars (h_2)	0.75 m
Height of camera (H)	6.00 m
Pitch angle ($-\theta$)	-10 °
The horizontal distance between RP and camera (L)	34.00 m

can be achieved through both manual alignment and automatic alignment. At the first start-up of the system, its user needs to manually align the positioning reference points in the virtual world with the corresponding points in the real world. Several buttons were designed to achieve the manual alignment, as shown in Fig. 6. The automatic alignment is activated when the manual alignment is finished. SLAM tracks the MR goggle based on image sequences taken by the goggle and inputs the movement into the visualisation module. Therefore, the displayed virtual world can keep the position all the time.

4. System testing and demonstration

A preliminary experiment was performed to validate the visual warning system in this study. The aim was to evaluate the feasibility of the proposed system for presenting hazard information. The hazard identification module was run on a server computer, and the real-time tracking module was run on a client computer. The development environment is shown in Table 2. An iPhone 12 mini was used to capture live streaming. As a wearable MR device, a Microsoft HoloLens 2 was chosen to run the visualisation module. The server software and MR application were developed using Unity3D version 2019.4.23f1c1.

The experiment was designed to determine the proposed system's feasibility in the real world, including effective ranges, tracking accuracy, and transmission latencies. Intersection over Union (IoU) is a commonly used indicator to evaluate object detection accuracy. As Shown in Fig. 9, IoU refers to the ratio of the overlap area to the union area between a ground-truth bounding box and the corresponding predicted bounding box. In this experiment, the concept of IoU was utilised to evaluate the accuracy of hazardous area visualisation. Since the accuracy of hazard visualisation is crucial to avoiding injury, strict IoU thresholds were chosen, i.e., above 90% as acceptable and above 95% as excellent. For a circular area of 5 m radius, the area of the red zone is 78.54 m^2. The error is acceptable when the distance between a displayed virtual object and the corresponding real object should be within 41.35 cm. If the distance is within 20.14 cm, the alignment is considered excellent.

A public place around Building 8 at RMIT University in Australia has been chosen to perform the experiment. Before the experiments began, the camera had been set at the height of 4.8 m. Its pitch angle had set to −25, as shown in Fig. 10 and Table 3. Then, the hazard identification and real-time tracking modules were run and tested in order. The position and pitch angle of the camera had inputted into the hazard identification module to align the virtual world in the system with the real world. The point of intersection between the camera's optical axis and the ground had determined as a Reference Point (RP). The RP in the

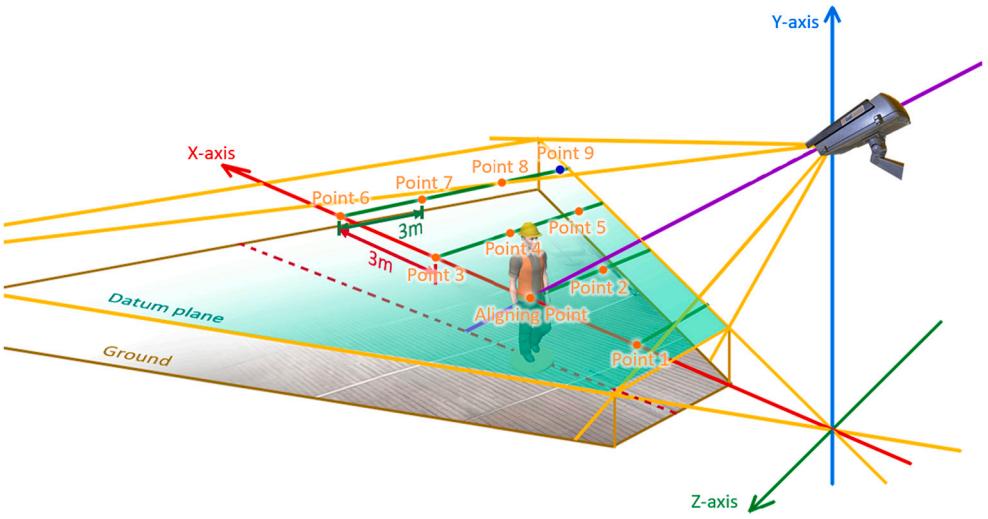


Fig. 11. The position of test points in Scenario 1.

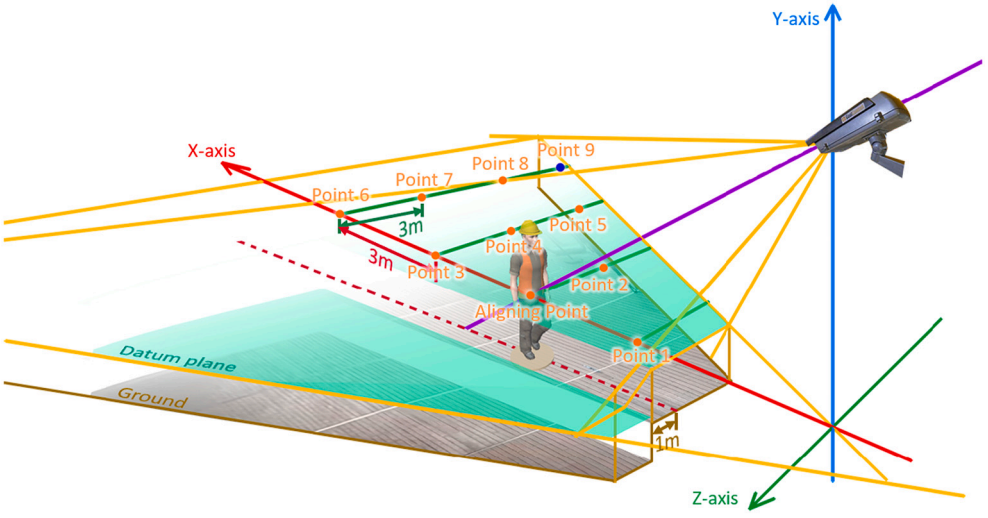


Fig. 12. The position of test points in Scenario 2.

virtual world had manually aligned with the RP in reality as accurately as possible to ensure that the DTs in the virtual world had overlapped exactly with the corresponding real-world object.

Three scenarios were designed to validate the proposed system. The hazardous situations in the scenarios include explosion, falling from heights, and struck-by accident. These hazardous situations involve tracking multiple objects in different complex 3D environments, which has requirements for robustness. The proposed system can track objects with no signal transceivers attached to them and collect context-aware information from BIM to generate hazard alerts related to workers' surrounding physical environment. These features make the proposed method more robust and accurate than conventional tracking and warning methods. The details of the tests are shown as follow:

4.1. Scenario 1: Explosion

Scenario 1 was designed to simulate two workers performing material moving tasks on a construction site. It aims to test the feasibility and accuracy of object tracking and displaying functions. A worker (the first participant) is responsible for moving an acetylene cylinder, and the other worker (the second participant) moves an oxygen cylinder. The safety regulation says the minimum distance between acetylene and oxygen cylinders is 5 m. Therefore, a red zone is within 5 m of the 2nd participant, and the acceptable error is 41.35 cm. Since the average human walking speed is 1.2 m per second, the yellow zone is within 11 m, and the blue zone is within 17 m. Based on the scenario, effective ranges, tracking accuracy, and transmission latencies of the proposed system were tested and recorded at several positions (point 1–9 in Fig. 11), considering the deviation of coordinates may occur on both the x and y axes.

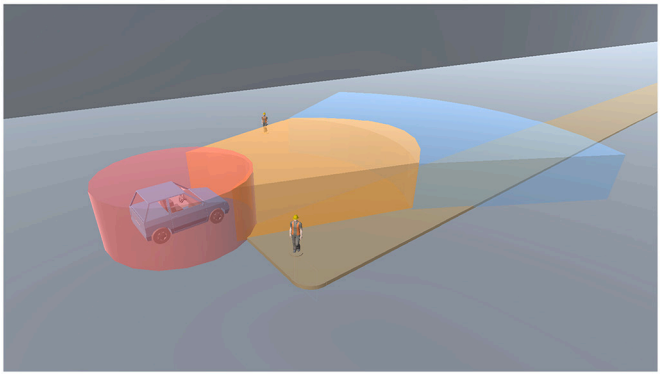
The experiment involves three steps. In step 1, the 2nd participant stands at the aligning point facing the camera. The 1st participant observing the 2nd participant uses the visual warning system to adjust his registration until the virtual 3D model of the 2nd participant and the real one is overlapped. Then, the 2nd participant stands at point 1 facing the camera. In step 2, the 1st participant marks the 3D model's position on the ground and records the network latency shown by the developed applications. The 2nd participant marks his own position on the ground. After that, the 2nd participant moves to the next point. In step 3, step 2 is repeated until the distance between the 3D model and the 2nd participant exceeds 41.35 cm, or the 2nd participant is out of the camera's view. The test was repeated for additional three times.

Table 4
Movement sequences of the participant.

Test no.	Movement sequence
Test 1	4–8–6 – out of viewing range – 7–5–2–3–1
Test 2	7–3 – out of viewing range – 8–6–5–4–2–1
Test 3	8–5–6–3 – 1–2 – out of viewing range – 7–4



(a)



(b)

Fig. 13. Identification of the hazardous situation: (a) shown in the object tracking program (b) shown in a VCS.

4.2. Scenario 2: Falling from heights

Scenario 2 was designed to simulate a worker (the participant in the test) approaching a pit without installed guardrails. It aims to test the accuracy of BIM geometry model registration and the feasibility of hazard warnings generated from BIM models. In addition, the accuracy of multiple objects tracking was also evaluated to validate the result in scenario 1. According to the safety regulation, guardrails should be set around a pit. However, the installation of guardrails needs time, which can cause a temporary hazard area. In Fig. 12, the lower part is the pit, and the edge is at $z = -1$. As a rule of thumb, the red zone is within 1 m, the yellow zone is within 2 m, and the blue zone is within 3 m. Based on the scenario, the alignment accuracies of BIM models and coordinates of the participants were recorded at points 1 to 8 in Fig. 12.

The experiment involves three steps. In step 1, the participant wearing HoloLens adjust his registration until both the virtual camera and alignment point are overlapped with the real ones. Then the participant stands at aligning point. In step 2, the participant walks to a random point or walks out of the camera viewing range, and the experimenter recorded the participant's position shown in the system. In step 3, step 2 is repeated until all the points have been visited. Then, measure the distance between aligning point shown in HoloLens and the real one. The test was repeated three times, and the movement sequences are shown in Table 4.

4.3. Scenario 3: Struck-by accident

Scenario 3 was designed to simulate that moving vehicles are approaching a worker (the participant in the test), which may cause a struck-by accident on a construction site. The test aims to evaluate the feasibility of tracking multiple objects on surfaces at different heights and providing suitable warnings. As shown in Fig. 13, cars were on the ground, and the participant was standing on a platform, which is 0.1 m above the ground. Thus, the datum planes are different for the cars and the participant. However, a height difference of 0.1 m cannot prevent

Table 5
Performance of the real-time tracking module in the proposed system.

Scenario	Accuracy	TPR	PPV	FNR	FDR
1	96.5%	97.0%	99.4%	3.0%	0.6%
2	96.7%	97.5%	99.2%	2.5%	0.8%
3	96.1%	97.0%	99.1%	3.0%	0.9%
Overall	96.4%	97.2%	99.2%	2.8%	0.8%

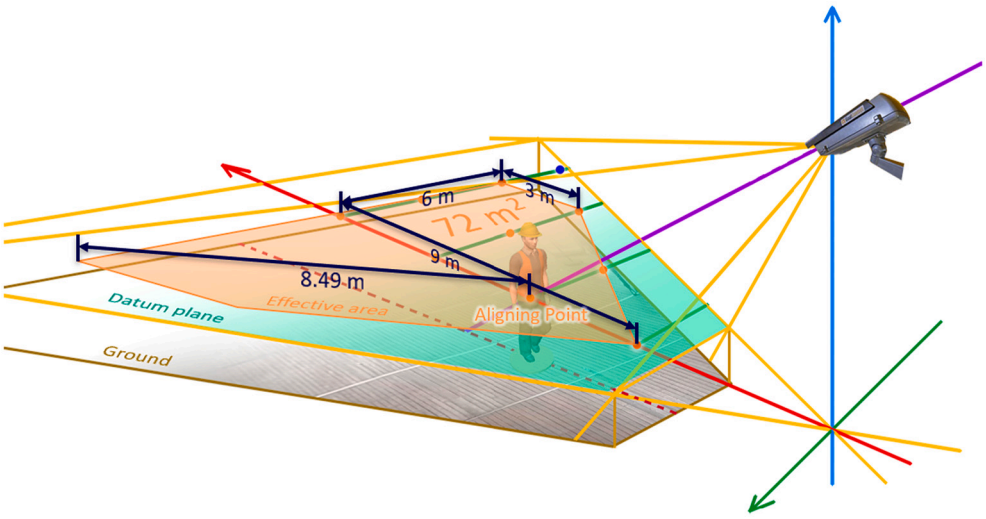
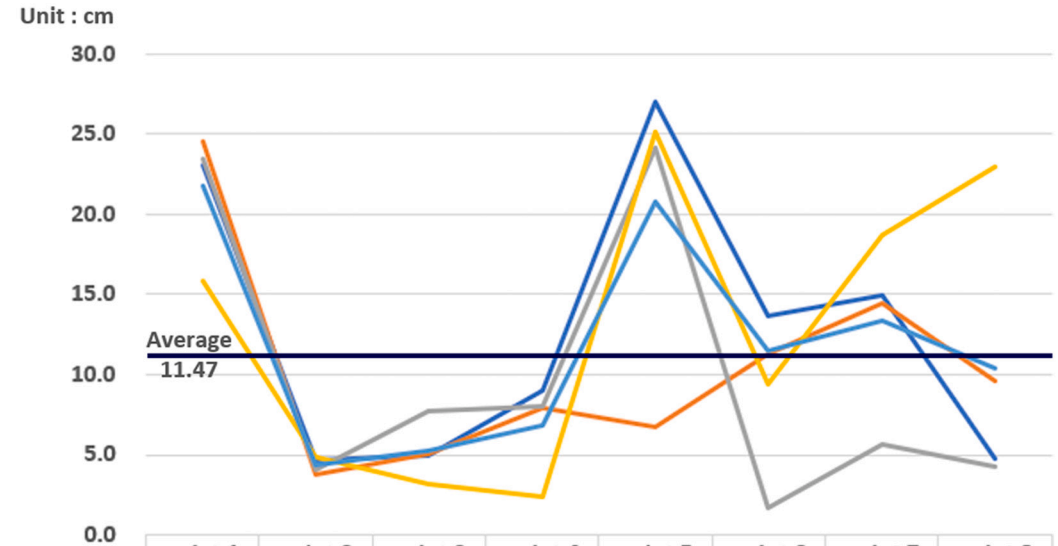


Fig. 14. The effective range of the real-time tracking module.

(a)



	point 1	point 2	point 3	point 4	point 5	point 6	point 7	point 8
test1	23.1	4.7	4.9	9.0	27.0	13.7	14.9	4.8
test2	24.6	3.8	5.1	7.9	6.8	11.3	14.4	9.7
test3	23.4	4.0	7.7	8.0	24.2	1.7	5.7	4.2
test4	15.8	4.8	3.1	2.4	25.2	9.4	18.7	23.0
average	21.7	4.3	5.2	6.9	20.8	11.5	13.4	10.4

(b)

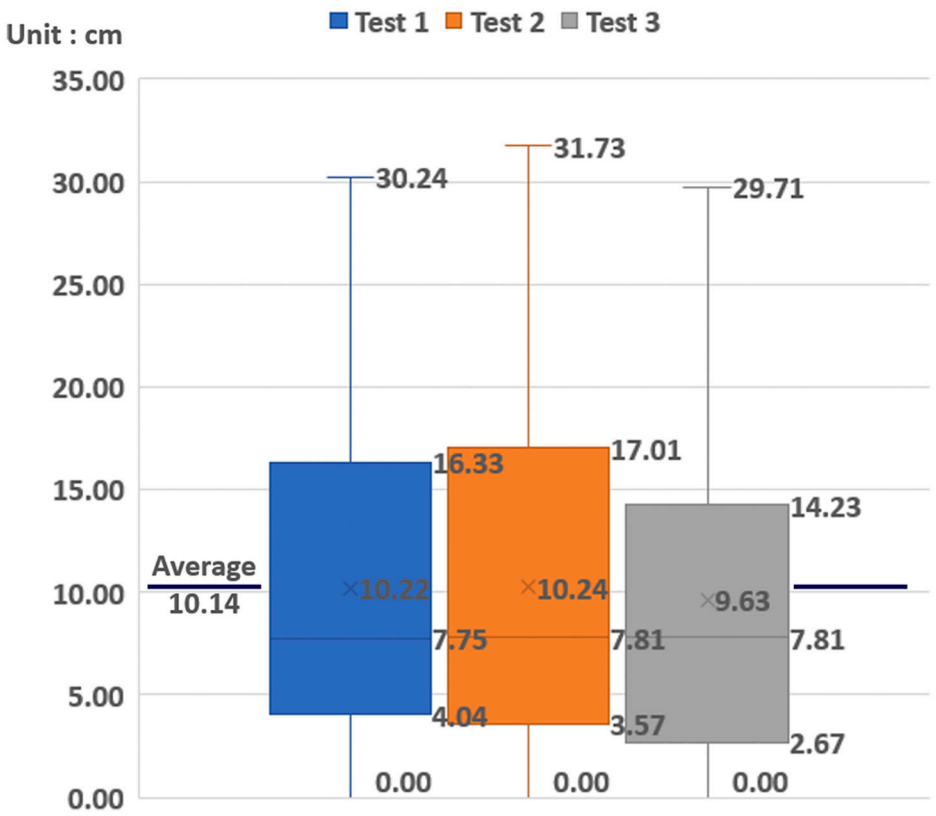


Fig. 15. The statistics of the distance between the real objects and their counterparts in the system: (a) Scenario 1 (b) Scenario 2.

struck-by accidents, so the proposed system should identify the hazard and provide warnings.

The experiment involves two steps. In step 1, the participant wearing HoloLens adjust his registration until both the virtual camera and alignment point are overlapped with the real ones. Then the participant stands at aligning point. In step 2, the participant and the experimenter record whether the hazard area is present at the proper position when a car is approaching.

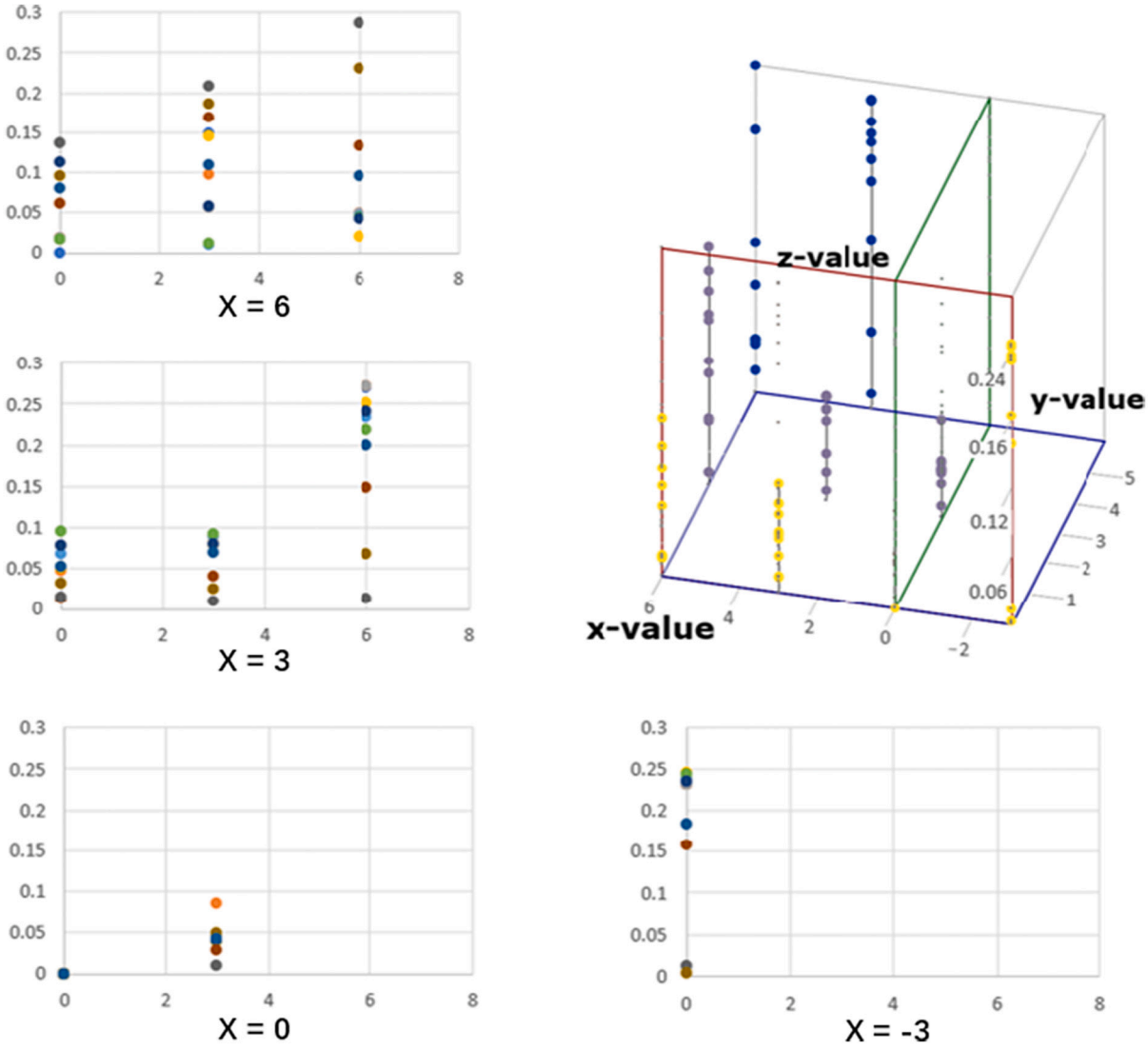


Fig. 16. The distance data between the real objects and their counterparts in the system at the different testing locations (unit: m).

5. Results and discussion

In scenario 1, an error of more than 41.35 cm occurred at point 9 (the point marked in blue in Fig. 14). Therefore, the effective range of the real-time identification module is within 8.49 m from the aligning point, and the effective area is more than 72 m^2 (area marked in yellow in Fig. 14) in a conservative calculation. Within this area, the IoU is greater than 90%. Using a local area network created by a mobile phone hotspot, the average latency of data transmission was 289.508 ms, and the maximum latency was 424 ms.

Two main functions should be evaluated in the proposed system. The first function is object detection, and the other one is information demonstration. When objects are correctly detected, it is called true-positive. If there is no object and the object detection function correctly identifies the situation, it is true-negative. The real-time tracking module may make errors in detecting objects. It may identify other objects as target objects (false-positive) or fail to identify the target objects (false-negative). There are five indicators to evaluate the proposed object detection function.

$$Accuracy = (TP + TN)/(TP + TN + FP + FN) \quad (5)$$

$$True\ Positive\ Rate\ (TPR) = (TP)/(TP + FN) \quad (6)$$

$$Positive\ Predictive\ Value\ (PPV) = (TP)/(TP + FP) \quad (7)$$

$$False\ Negative\ Rate\ (FNR) = (FN)/(TP + FN) \quad (8)$$

$$False\ Discovery\ Rate\ (FDR) = (FP)/(TP + FP) \quad (9)$$

where TP = True-Positive, FN = False-Negative, FP = False-Positive, and TN = True-Negative.

The result is shown in Table 5. The object detection accuracy is more than 96.5%, which is good enough to provide hazard information to on-site workers. In the view of system users, it has been observed that the false-negative cases have little effect on the visualisation, but false-positive cases are more significant. Due to the nature of object tracking algorithms, coordinates provided by the algorithms change all the time, even if a target object is not moving. It causes the shaking of the corresponding DT. Therefore, an anti-shake script was developed and applied to avoid the shaking of DTs. The side-effort is that the DTs of tracked objects disappear after three additional frames (about 0.4 s) in the visualisation module. Therefore, the failure of object detection in a single frame does not affect presenting hazard information. However, if an object is identified as two objects, two virtual models are shown in MR goggles simultaneously for four frames.

For the evaluation of information demonstration, the experimental

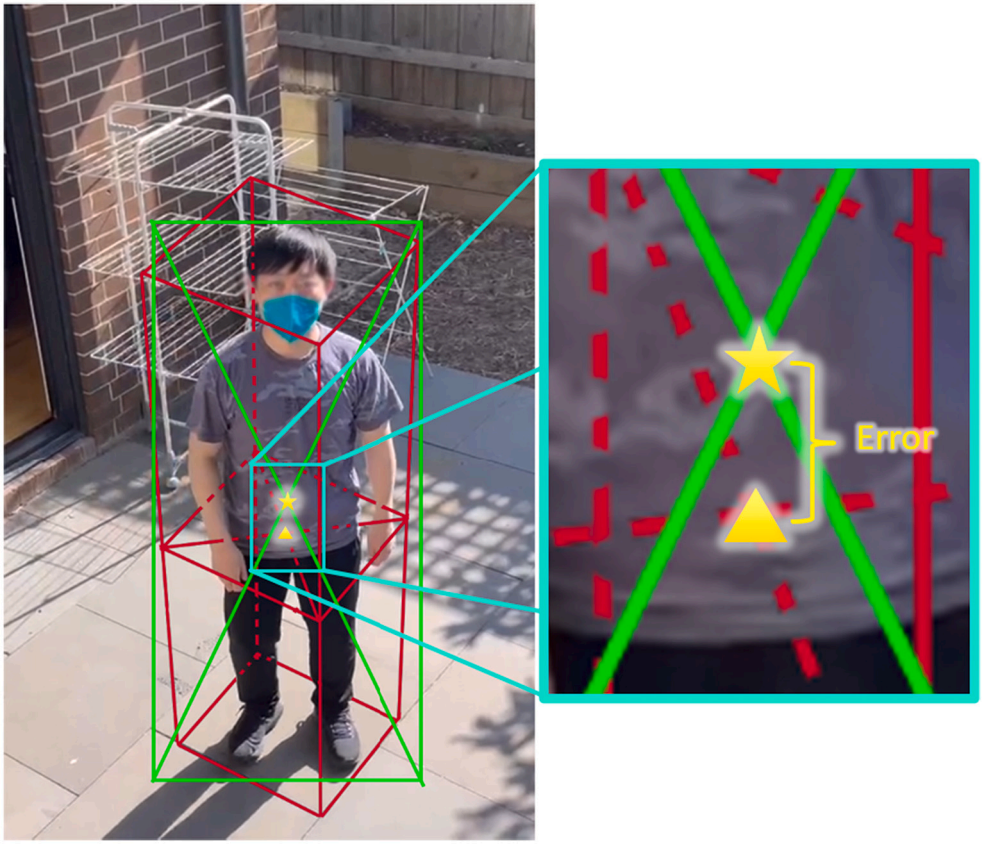


Fig. 17. The error caused by the difference between the centroid of a tracked object and the centre point of the recognition boundary box.



Fig. 18. Hazardous area identified in scenario 3.

results show that the average distance from a displayed 3D model to the corresponding real object is 11.47 cm in Scenario 1 and 10.14 cm in Scenario 2, as shown in Fig. 15. In more than 75% of cases, the distance is less than 17.01 cm. Besides, the position of the geometry model from BIM files is stable. The deviation of the results of all three repeat tests is less than 1.5 cm (0.9, 0.4, 1.3, respectively).

By comparing the errors at different positions in both Scenario 1 and 2, the results show the same trend. The recorded errors at different positions in Scenario 2 are summarised in Fig. 16, which is helpful to analyse the causes of the errors. The figure uses the aligning point as the origin of the coordinates system. X and Y values refer to the positions of testing points in the real world. If the X value of an object is negative, the object's position is between the aligning point and the camera. The Z values are the recorded errors, reflecting information presenting

accuracy.

The experimental results show that the object tracking algorithm completely determines the accuracy of the visualisation. The maximum errors become larger with the increase of x and y values. This phenomenon can be explained by the fact that if the testing point is farther away from the camera, the actual distance between two adjacent pixels is larger. Besides, the error becomes larger as a tracked object gets closer to the camera. A phenomenon can be found that when the x value is −3, the boundary boxes shake worse than at other positions, which increases the error to some extent. Besides, the top half of the object is significantly larger than the bottom half at this position. This is because when an object is approaching the camera, the phenomenon of perspective is becoming more obvious, as shown in Fig. 17. The centre point (marked as a star) of the recognition boundary box of a tracked object has moved away from the object's centroid (marked as a triangle). Thus, there is a significant increase in error. The lack of a method to eliminate the phenomenon of perspective is a current limitation of the proposed system. In addition, lens distortion is another possible reason to cause errors, which has not been taken into account. It is another limitation of the proposed system.

In Scenario 3, the proposed system successfully detected the spatial position of objects on different planes and identified potential hazards. The hazardous areas, including red, yellow, and blue zones, were automatically moved forward of vehicles according to the latency and speed of tracked objects to eliminate the demonstration error caused by latency. The yellow and blue zones are extended according to the calculated speed of the vehicle (e.g., 27 m for a 20 km/h car). The results show that for 84 cars out of 93 cars, the hazardous areas are well identified and displayed, as shown in Fig. 18. The other cars were turning at the junction, which caused the red zone cannot cover them very well. However, the yellow and blue zones can still warn system users in advance to let them know the hazard sources are approaching. In order to cover all the hazardous areas, the hazard identification function should be further developed to expand the identified hazardous areas based on the movement trend of tracked objects.

In the testing scenarios, participants have a tendency to move away from the displayed hazardous areas. The phenomenon indicates the proposed system has the potential to reinforce workers' safety behaviour. In the future, more experiments should be conducted to evaluate the influence of the displayed hazardous areas on the system users, especially their behaviours. The influence should be tested and analysed from the aspects of engineering psychology and cognitive ergonomics. The indicates should include participants' ability to recognise hazardous situations, participants' response time to specific hazards, and the duration of participants staying in each hazardous zone. Besides, the NASA Task Load Index should also be included to test whether and to what extent the additional information provided by MR devices increases users' workload.

6. Conclusion

Presenting hazard information is helpful to support on-site workers in decision-making. However, conventional safety management does not adequately take advantage of emerging technologies to prevent workers from construction accidents. This study proposes a visual warning system to provide hazard information to workers, allowing workers to be aware of surrounding hazards and reassess risks. By integrating BIM, CV, and DL technologies, a VCS were developed in the system. The VCS simulates construction activities, identifies hazards, and generates hazard information based on BIM models and real-time information obtained from on-site cameras. The proposed system is able to present the hazard information to workers through wearable MR devices, letting them take proactive actions to avoid injuries. Three outdoor tests have been completed to verify the feasibility of the proposed system. According to the results, the proposed system performs well in merging virtual objects with the real world and presenting hazard information. The effective area of the visualisation is more than 72 m^2 with a delay of no more than 424 ms when the camera is set at 5.68 m height with an -25° pitch angle. The hazard identification accuracy is more than 96.1%, and the average hazardous area presentation accuracy is 11.47 cm.

Nevertheless, some issues in this study need to be addressed in the future. Firstly, the function to eliminate lens distortion and the phenomenon of perspective should be developed. Besides, an image-based prediction function will also be introduced into the proposed visual warning system. The hazardous areas caused by high-speed moving objects will be identified more accurately. More IoT devices will be embedded in the system to cover more hazard types. Moreover, the alignment of a virtual world with the real world needs to be done manually by users. In the future, the system will utilise spatial anchors to share spatial coordinates between different devices, enabling more accurate and automated alignment. In addition, the application of AR and MR has been proven to be effective in enhancing worker performance, but the underlying rationale is not clear yet. Further theoretical research is needed to clarify the impact of advanced visualisation technology on workers' behaviours based on engineering psychology and cognitive ergonomics.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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